MICHELE MAIULLARI

Aeronautical & Mechanical Engineer

CONTACT

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DETAILS

Date of birth: 27.03.2001
Immediate availability for international travel & relocation
EU driving licence – category B
Italian citizen · valid passport

SOFTWARE & TOOLS

CAD / PLM
CATIA · 3DEXPERIENCE · Autodesk Inventor · AutoCAD · FreeCAD

FEA & Simulation
Ansys (structural, fatigue, loads) · MATLAB · Simulink

MBSE
CATIA Magic System Architect (SysML)

Programming
Python · Fortran · Arduino

AI / Machine Learning
LLMs · open-source models · performance & weight evaluation

Dev & OS
Git / GitHub · Linux · Windows · macOS

Office
Excel · Word · PowerPoint

LANGUAGES

- Italian — Native
- English — Proficient
- German — Basic (in course)
- French — Basic (in course)

SOFT SKILLS

- Cross-functional, multidisciplinary coordination, teamwork
- Stakeholder management
- Technical & requirements documentation
- Organization & time management

CERTIFICATIONS

- Leonardo Aerotech Academy
- AMTECH – Additive Manufacturing Technologies
- Training Course for Workers

PROFILE

Aeronautical (MSc) and mechanical (BSc) engineer with industry experience at Leonardo Aerostructures and SKF, hands-on across the full development cycle of UAS components — 3D CAD design, structural validation, manufacturing, assembly and testing. Applies MBSE, requirements management and V&V from concept to qualification; currently leading an early-stage AI project for automated airfoil generation.

CORE SKILLS

- **Systems Engineering:** MBSE & SysML · requirements management · trade studies · V&V
- **Structural Analysis:** FEA/FEM · fatigue & load evaluation · root-cause analysis
- **Aircraft Design & Aerodynamics:** CFD · flight mechanics · flight simulation · UAS/UAV design
- **Manufacturing & Quality:** additive manufacturing · rapid prototyping · series production & quality control · HW–SW integration

EXPERIENCE

Engineering Intern — Aerotech Academy
Oct 2025 – Jul 2026

Leonardo S.p.A · Aerostructures Business Unit — Grottaglie, Italy

Coursework: Aerospace · Flight Mechanics · Structures · Advanced Materials · Sustainable Manufacturing · Digital Systems

- Designed a wing and a quick-release wing–fuselage coupling for MALE-ISR UAS platforms, in collaboration with the Aerostructures R&D department.
- Covered conceptual, preliminary and detailed 3D CAD modelling, structural validation and stress analysis.
- Supported the manufacturing, assembly and testing phases.

Additive Manufacturing Training — AMTECH
Mar 2025

IEC Tehnopolis — Nikšić, Montenegro

Coursework: AM Applications · Business Models · Advanced Industry Seminar by Meltio

- Applied AM theory to real production processes: reverse engineering and prototype development via advanced 3D scanning and printing technologies at TechLab.

Engineering Intern — Quality & Vibration Analysis
Mar – Jul 2023

SKF S.p.A — Modugno, Italy

- Performed quality control on single-row deep-groove ball bearings for electric-automotive applications, classifying rejects by noise frequency band (NVH Analysis).
- Ran vibration analysis and root-cause troubleshooting on non-conforming units, collaborating with production to optimize processes and reduce scrap.

EDUCATION

MSc Aeronautical Engineering — Politecnico di Bari
Nov 2023 – Oct 2026

Coursework: Fluid Dynamics · Energy Systems · Aeronautical Materials & Manufacturing · Design & Simulation

- **Thesis (with Leonardo S.p.A):** *"Design and Testing of Lightweight Aeronautical Systems, UAVs and UASs, Built Using Rapid Assembly and Deployment Techniques."*

Erasmus Exchange — RWTH Aachen University
Mar – Oct 2025

- Non-linear FEM · fatigue of structures · fatigue design of lightweight structures · model-based product development.

BSc Mechanical Engineering — Politecnico di Bari
Oct 2020 – Nov 2023

Coursework: Applied & Structural Mechanics · Thermo-Fluids & Energy Systems · Manufacturing & Materials · Design · Mechanical & Thermal Measurements

- **Thesis (with SKF S.p.A):** *"Vibrational Analysis of Single-Row Deep-Groove Ball Bearings for Electric Automotive Applications."*

International Exchange — Italian High School Terra Santa
Sep – Oct 2019

- Haifa, Israel · Immersive academic and cultural exchange

SIDE PROJECTS

AI-Based Airfoil Generation Platform — Project Lead
Jan 2026 – Present

- Leading the development of an AI-based aeronautical software system that automatically generates, optimizes and validates airfoils and control surfaces; driving business development and technical validation with key industry stakeholders.

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